

Validator Agent: Two-Month Research Plan

Goal: deliver a reproducible inference-time validator and a complete research-paper draft within eight weeks. The validator checks scientific conclusions produced by an upstream agent, separates the support in its original citations from independently retrieved evidence, and explains uncertainty to a non-expert reader.

Planning assumptions: one lead researcher working 18 hours per week, plus 16 hours of contingency across the project (160 hours total). Independent scientific annotation requires additional collaborator time. The schedule uses relative days 1-56; start on any convenient Monday. All experiment sizes, thresholds and costs below are proposed planning choices, not completed results.

Recommended scope

Use biomedical research abstracts for the first version because SciFact provides scientific claims, evidence and rationales. Keep the task about interpreting research evidence, not giving treatment advice. Start with a frozen local corpus and replaceable local model endpoints. A second model family tests portability; model-agnostic architecture does not by itself establish generalization.^{6,13}

By the end	Required output
Week 2	Working same-evidence baseline, corpus index, schemas and locked split manifest.
Week 4	Independent retrieval, staged judgment, evidence cards and annotation pilot.
Week 6	Locked main experiments, cost analysis, calibrated outputs and figure drafts.
Week 8	Complete paper, limitations, reproducibility package and exploratory user-study results, if feasible.

How to use this document

Pages 2-8 define the research and implementation. Pages 9-14 specify data, comparisons, metrics, software tests and the human study. Pages 15-22 give a day-by-day eight-week schedule. Pages 23-24 cover resources and the submission package; pages 25-26 contain the source register.

Definition of completion: an honest, reproducible paper-ready package. A positive result, publication acceptance, or a statistically conclusive human study cannot be guaranteed in two months.

6. Wadden et al.. Fact or Fiction: Verifying Scientific Claims. EMNLP, 2020.

13. Allen Institute for AI. SciFact data, models and evaluation code. repository, accessed 12 September 2026.

1. Positioning and defensible novelty

The original framing needs narrowing. Atomic decomposition, blinded checking, search-based verification and readable explanations all have substantial prior art. The proposed contribution is the design and evaluation of their specific combination under controlled evidence access.

Prior work	Established overlap	Consequence for this paper
MARCH [1]	Solver, Proposer and isolated Checker; shared-policy reinforcement learning.	Borrow information separation; do not claim to invent blinded checking.
Blame localization [2]	Diagnoses faithfulness and citation errors across agent traces.	Evaluate final-user evidence judgments; trace localization is supplementary.
CoVe [3]	Answers verification questions independently of the draft.	Include a factored question-checking control.
SAFE [4]	Decomposes long responses and verifies facts using search.	Fresh retrieval alone is not a new contribution.
SciClaims [5]	Scientific extraction, retrieval, verdicts and natural-language justifications.	A standalone interface or explanations alone are insufficient.
RT4CHART [9]	Claim-level context verification and evidence-linked spans.	Compare context faithfulness separately from broader evidence support.

Research questions to preregister

RQ1: At a matched inference budget, does independent retrieval plus staged evidence-first judgment reduce false endorsement compared with same-evidence checking?

RQ2: Can the system distinguish missing evidence from unresolved evidence and support useful abstention?

RQ3: Do evidence-linked explanations improve non-experts' judgments, including when the validator is wrong?

RQ4: How do accuracy, coverage, latency and cost change across generator/checker families?

Use wording such as "we evaluate a staged independent-evidence validator for agent-generated scientific conclusions." Avoid "the first" or "nobody has measured human trust": human trust calibration already has experimental literature.¹¹

1. Li et al.. MARCH: Multi-Agent Reinforced Self-Check for LLM Hallucination. 2026, v1.

2. Hirsch et al.. Who is the Agent to Blame? Localizing Faithfulness and Citation Mistakes in Agentic Deep Research. 2026, v1.

3. Dhuliawala et al.. Chain-of-Verification Reduces Hallucination in Large Language Models. 2023 preprint; 2024 publication.

4. Wei et al.. Long-form factuality in large language models. 2024.

5. Ortega and Gomez-Perez. SciClaims: An End-to-End Generative System for Biomedical Claim Analysis. EMNLP System Demonstrations, 2025.

9. Yu et al.. Retromorphic Testing with Hierarchical Verification for Hallucination Detection in RAG. 2026, v2.

11. Zhang, Liao and Bellamy. Effect of Confidence and Explanation on Accuracy and Trust Calibration in AI-Assisted Decision Making. FAT*, 2020.

2. Paper structure following MARCH

MARCH follows Introduction, Methodology, Experiment, Related Work and Conclusion, with prompts and evaluation details in appendices. Use that organizational pattern, replacing training-specific material with inference-system design and evaluation.¹

Paper section	What to write and prepare	Draft timing
Abstract	Problem; proposed mechanism; evaluation setting; measured results; limitation. Write numbers only after experiments.	W1 skeleton; W8 final
1 Introduction	A concrete failure example; scope; research gap; three testable contributions.	W1, revise W7
2.1 Problem formulation	Inputs, corpus boundary, evidence scopes, labels, abstention and research questions.	W1-W2
2.2 System design	Information-flow diagram; independent collection; two judgment stages; dependency checks.	W3-W4
2.3 Implementation	Schemas, retrieval settings, model revisions, prompts, budgets, stopping and failure handling.	W2-W4
3.1 Experimental setup	Datasets, split policy, annotation, baselines, compute and preregistered outcomes.	W2-W5
3.2 Main verification results	Native benchmark scores, four-way results, false endorsement and intervals.	W6
3.3 Agentic conclusions	Natural errors, multi-paper reasoning, corpus-relative evidence and transfer.	W6-W7
3.4 Ablations	Evidence access x information separation; budget controls; oracle evidence.	W6
3.5 Further analysis	Calibration, selective risk, cost, failure taxonomy and human-study pilot.	W7
4 Related Work	Mechanism-by-mechanism comparison, including the closest scientific and search verifiers.	W1-W2, refresh W8
5 Conclusion	Answer each RQ at the strength supported by results; unresolved problems.	W8
Limitations / ethics	Corpus coverage, label ambiguity, annotation expertise, pilot power, misuse and access limits.	Start W1
Appendices A-E	Prompts / schemas; data card / rubric; run configs; detailed results; study materials.	Continuously

Working allocation: roughly 8-10 main-text pages plus references and appendices. This is a drafting target, not a venue rule. Select a venue in week 1 and verify its current length, anonymity, ethics and artifact requirements before submission.

1. Li et al.. MARCH: Multi-Agent Reinforced Self-Check for LLM Hallucination. 2026, v1.

3. Architecture and information boundaries

Implement logical roles as isolated functions with typed inputs. Several roles may call the same model in separate stateless requests. Separate roles do not guarantee statistically independent errors, so test model-family dependence explicitly.

Component	Allowed inputs	Output and boundary
Sensemaking loop	Question, initial retrieval results	Frozen conclusion and original citations D0; preserve its trace.
Proposer	Question and frozen conclusion	Atomic claims, neutral questions, asserted answers and source spans.
Independent gatherer	Neutral question, scope, corpus	Fresh evidence D1. Hide conclusion, asserted answer, original citations and generator identity.
Evidence reader	Neutral question and shuffled D1	Evidence-only answer and cited spans; seal this record before revealing the claim.
Judge	Atomic claim, sealed record, D1	Independent-evidence label and uncertainty; no full report required.
Citation auditor	Atomic claim and D0	Separate original-citation label, spans and citation defects.
Reconciler	Two judgments and claim dependencies	Explain differences; assess combined evidence; flag invalid inference links.
Renderer	Validated structured record	Plain-language claim cards and cautious report summary; no new factual claims.

Execution and stopping

Generate the report once, freeze it, and validate without an automatic rewrite loop in the main study. Start with at most 8 claims per report, 2 retrieval rounds per claim, 3 query forms per round and 8 evidence passages in the judge context. Treat these as development defaults; freeze them before testing. If a report exceeds the claim budget, mark partial coverage and list unchecked spans rather than silently dropping them.

Round 2 runs only for unresolved questions, scope mismatch or contradictory evidence. Stop on exhausted budget, no new eligible documents, or a complete bounded evidence record. Persist all intermediate artifacts. A timeout creates an execution-failure status, never an evidence verdict.

Why two judgment stages matter

The final judge must see the claim to classify it. The blind component is the preceding evidence-only reading stage. Neutral questions still reveal the topic and may leak assumptions; audit this leakage rather than describing the entire validator as perfectly blind.

4. Implementation contracts and repository

Use Python, typed request/response validation, a command-line runner, a local retrieval index and a replaceable inference client. Add a small FastAPI endpoint only after the batch runner is reliable. Use an HTML template for the study interface; a large frontend or agent framework is unnecessary for the research question.

Record	Required fields
Run	run_id, question_id, split, seed, model_id/revision, prompt_hash, corpus_hash, config_hash, timestamps, tokens, status
Claim	claim_id, report_id, exact source span, normalized claim, subject, relation, object, population, comparator, time, units, modality, neutral question, asserted answer, dependencies
Evidence	doc_id, DOI/PMID if present, snapshot hash, text span and offsets, query, rank, retrieval round, original/independent provenance, deduplication group, access scope
Judgment	claim_id, evidence scope, label, rationale codes, cited span IDs, raw scores, calibrated probabilities if valid, execution status, uncertainty reasons
Report verdict	claim coverage, central claim outcomes, inference-link status, conflicts between D0 and D1, display explanation, limitations

Suggested file responsibilities

src/validator/: schemas.py, decompose.py, retrieve.py, evidence_reader.py, judge.py, reconcile.py, calibrate.py, render.py, runner.py.

src/evaluation/: adapters.py, metrics.py, bootstrap.py, study_analysis.py.

configs/: model, corpus and experiment YAML files.

prompts/: versioned role templates.

data/: manifests and permitted data; keep source text separate from labels.

tests/: contract, isolation and regression cases.

paper/: section drafts, references, figures and appendices.

Reproducibility contract

Each completed run emits a machine-readable results file and immutable trace. Cache by normalized input, model revision, prompt, corpus, seed and retrieval settings. Include split in evaluation metadata; caches must never expose labels or private evaluator pools to inference. Restart at the last completed component without mixing old and new configurations.

One command should index a supplied corpus; one should validate frozen reports; one should compute tables from saved predictions. Ship a tiny licensed example so reproduction can be checked without a GPU or live search.

5. Claim decomposition and retrieval

Atomic scientific claims

Preserve quantities, confidence language, population, experimental setting and causal scope. Split "X improves A and B" into two claims; retain the relationship between them. Do not turn "is associated with" into "causes," drop "in mice," or remove a comparator. Mark opinions and recommendations separately. Map every claim back to exact report text and audit coverage against human-extracted claims.

Example: "Across two studies, compound X caused a 20% improvement in memory in older adults." Extract study-count, causal-effect, effect-size and population assertions, plus the synthesis relation. A paper supporting one component does not validate the combined sentence. This is a synthetic example, not a biomedical finding.

Frozen retrieval protocol

Begin with BM25 over the SciFact abstract corpus. On development data only, compare BM25 with a dense retriever and reciprocal-rank fusion. A proposed fusion score is the sum of $1/(60 + \text{rank})$ across lists. Record the chosen encoder, tokenizer, index settings and revision rather than selecting a model by brand reputation.

For each neutral question, create an open inquiry, a scope/measurement query and a limitations-or-null-results query. Retrieve top 20 candidates per query, deduplicate, rerank the union and select up to 8 passages. Keep one document's adjacent context together when needed to resolve negation or units. Log missing metadata rather than inferring study design from an abstract.

Evidence independence is measurable

D1 may legitimately rediscover papers in D0. Do not suppress overlap in the main system: that would discard potentially essential evidence. Report document overlap, unique evidence yield and supporting/refuting evidence yield. Use an exclusion-of-D0 condition only as a labeled diagnostic. Deduplicate preprint/publication versions and repeated snippets; two copies are not two independent studies.

The two experimental settings are different: a frozen corpus measures reproducible search behavior, while live retrieval measures deployment behavior. Keep live-search results as a dated stretch experiment with archived permitted content, not a moving source of main-test labels.

6. Four labels, three evidence scopes

A verdict concerns the stated claim relative to a specified evidence collection and search budget. It is not an unrestricted declaration of scientific truth. The following is a proposed annotation policy to refine in a blinded pilot.

Label	Operational rule	Example of valid wording
Supported	Relevant evidence establishes the exact scoped assertion and no unresolved material counterevidence remains in the reviewed pool.	The reviewed study supports this effect in the tested population.
Contradicted	Relevant evidence establishes an incompatible assertion; apparent conflicts have been resolved by scope or relevance.	The source reports a decrease, while the claim says an increase.
Unaddressed	No reviewed relevant passage answers the claim under the stated corpus and retrieval budget.	We did not find evidence addressing this claim in the searched collection.
Underdetermined	Relevant evidence exists, but disagreement, imprecision, incomplete measurement or an unsupported inference prevents resolution.	The studies address the question but do not establish a clear direction.

Decision order and reason codes

First check execution and claim validity. Then assess relevance. If no passage addresses the claim, use unaddressed. If relevant evidence is materially conflicting or inconclusive, use underdetermined. Otherwise apply support or contradiction only at the preserved scope. A non-significant result is not automatically proof of no effect. A null-result claim requires its own evidence assessment.

Suggested underdetermined reasons: conflicting results, wide uncertainty, missing comparator, causal overreach, population transfer, incomplete full-text access, unsupported synthesis. Use access-limited and search-limited flags separately. Low model confidence is epistemic uncertainty about the assigned label; it does not redefine that label.

Keep scope-specific outputs

D0: Is the original citation set faithful to the claim?

D1: What does independent retrieval establish?

D0 union D1: What can be said after reconciliation? A claim can be unsupported by its citation yet supported elsewhere. Report both outcomes; do not "repair" citation faithfulness using outside evidence.

For multi-hop conclusions, label inference links separately: valid, invalid or unresolved. Supported premises do not prove a causal or transitive conclusion. Report per-claim coverage and central-claim status; avoid averaging away a critical unsupported link.

7. Calibration and reader-facing output

Distinguish three quantities: evidence support, the validator's probability of assigning the correct label, and the reader's belief. Calibration evaluates a defined prediction against held-out labels; an LLM-generated percentage is not automatically calibrated.¹⁰

Practical calibration procedure

Prefer a constrained four-label classifier with accessible class scores. Fit one temperature on the calibration split, minimizing multiclass negative log-likelihood. If the endpoint exposes no appropriate scores, use a clearly documented raw-score extraction method and fit a small regularized calibrator using development choices fixed in advance. If data or score quality is insufficient, show qualitative uncertainty and report that numerical calibration remains unvalidated.

Fit no parameters on test data. Keep public-benchmark and agentic-report calibration separate; their distributions differ. A roughly 100-claim agentic calibration set is small, so report wide intervals and avoid strong subgroup calibration claims. Compare raw and calibrated scores and include a constant class-frequency predictor as a sanity baseline.

Abstention policy

On calibration data, choose a threshold for the supported probability to seek a low false-endorsement rate, then measure achieved risk and coverage on test data. Do not promise a 5% risk bound from a small sample. A scientific "underdetermined" label and a model "defer to reviewer" action are separate fields. If probabilities are unreliable, retain deferral based on explicit missing-evidence or conflict flags.

Claim card template

Claim: exact, readable assertion.

Evidence verdict: label and a one-sentence explanation.

Why: linked supporting/refuting excerpts with study scope.

What is uncertain: conflict, missing information or an inference gap.

Search scope: corpus/date, abstract-only status and reviewed document count.

Original citations: adequate, inadequate or unverifiable.

Next useful check: the specific missing evidence, not generic advice.

Generate prose only from the structured record, then verify each explanatory assertion against stored spans. Never invent a missing control group, common lab affiliation or study independence. Use "not reported in the available abstract" when that is all the evidence supports. Evaluate readability and factual faithfulness separately.

10. Guo et al.. On Calibration of Modern Neural Networks. ICML, 2017.

8. Dataset and benchmark portfolio

Use two core datasets and one small transfer check. Expand only after the core experiments run. Official labels and custom labels must remain distinct.

Track	Planned use	Protocol and limitation
SciFact [6,13] - core	Scientific retrieval, evidence selection and native verification.	Use released training data for development/calibration; freeze public development split as a local held-out evaluation if official test labels are inaccessible. Name it accurately.
Agentic set - core	80 reports from 40 questions, two generator families; target 4-6 claims per report.	10 question families for development, 10 calibration, 20 test: 20/20/40 reports. Human labels required; small test set supports preliminary conclusions.
RAGTruth [8] - transfer	200 preselected held-out responses, stratified by task/model where available.	Keep source contexts fixed. Evaluate response-level hallucination and optional span metrics using native annotations. No outside-evidence relabeling.
SciFact-Open [7] - stretch	Open-domain retrieval stress test.	Larger corpus and incomplete evidence judgments require care: unjudged documents are not automatically negatives.
Missci [14] - stretch	Scientific reasoning and misrepresentation cases.	Use published task definitions; fallacy reconstruction is not identical to four-way claim labeling.

SciFact split discipline

Reserve a fixed calibration subset from the released training split before prompt tuning; for example 100 claims if available after grouping. Use remaining training items for development. Preserve official scorer inputs and outputs. Treat the public development split as held-out only within this project, never call it an untouched community test set. If official test evaluation is accessible, report it separately.

Report native support/refute/evidence metrics. Three-way insufficient-information labels do not supply gold four-way underdetermined labels. For a supplementary coarse mapping, collapse unaddressed and underdetermined into insufficient information and label that comparison clearly. Evaluate the distinction itself only on human-annotated custom data.

Data manifest

Record dataset version, retrieval date, license/access terms, counts, excluded IDs, grouping rules, hashes and provenance. Group duplicate claims, paraphrases, underlying question families and derivative reports in one split. Audit cross-benchmark overlaps; report unavoidable corpus overlap and possible model pretraining contamination.

6. Wadden et al.. Fact or Fiction: Verifying Scientific Claims. EMNLP, 2020.

7. Wadden et al.. SciFact-Open: Towards open-domain scientific claim verification. 2022.

8. Niu et al.. RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models. ACL, 2024.

13. Allen Institute for AI. SciFact data, models and evaluation code. repository, accessed 12 September 2026.

14. Glockner et al.. Missci: Reconstructing Fallacies in Misrepresented Science. ACL, 2024.

9. Agentic benchmark and gold annotation

Produce realistic reports

Write 40 research questions requiring evidence from at least two documents or a comparison of scope, methods or outcomes. Use a fixed two-round sensemaking loop: plan subquestions, retrieve, extract source notes and synthesize a 150-250-word conclusion with citations. Run two generator families with the same corpus and budget. Keep model-produced errors intact and preserve every trace.

Split question families before report generation. Human annotators extract reference claims independently; validator-generated claims are not the gold inventory. Mark each source span, centrality and inference dependency. Verify that multi-hop items actually require multiple pieces of evidence instead of merely mentioning two papers.

Independent reference evidence

For each claim, pool D0 and retrieval results from all frozen compared methods, anonymize provenance, then allow an annotator search pass over the same corpus. Pooling is evaluator-only and happens after predictions are locked; systems never receive competitors' evidence. Annotate relevant passages, exact scope and evidence sufficiency. This reduces, but does not eliminate, incomplete-gold bias.

Assign D0-only and pooled-reference labels separately, with a third field for inference-link validity. Store reasons, spans and unresolved disputes. Do not let the tested judge assign its own gold labels. The absence of a found source establishes only corpus-relative unaddressed status.

Annotation workload and agreement

Target 400 claims from 80 reports. At 4 minutes per claim per annotator, double annotation takes about 53 hours, plus 8-12 hours of adjudication and evidence searching. Difficult scientific claims may take much longer; time a 20-claim pilot. Recruit two scientifically literate annotators and an adjudicator, with relevant biomedical expertise for difficult cases. These hours are additional to the lead's 160-hour plan.

Double-label the pilot, revise the rubric, then freeze it. Report raw agreement, nominal Krippendorff's alpha, class counts and a disagreement matrix before adjudication. Do not impose an agreement cutoff as a universal scientific standard. If the four-label boundary stays unreliable, report that finding and prioritize coarse-label performance.

Separate robustness set

Create a small labeled diagnostic set of 40 controlled errors: numbers/units, negation, population, causal inflation and citation substitution. Keep it separate from naturally generated errors. Preserve its seed families in the same split; do not advertise injected errors as natural prevalence.

10. Baselines and ablation matrix

Use identical frozen reports, claim inventories where appropriate, corpus snapshots and scoring code. All baseline adaptations must be named; an inspired prompt is not a full reproduction of a trained method.

ID	Condition	Purpose
B0	Direct judge sees report + D0.	Minimum same-citation LLM-judge baseline.
B1	Atomic claims + D0; judge sees asserted answers.	Isolate decomposition from whole-report judging.
B2	Neutral questions + isolated reader + D0; then compare.	MARCH/CoVe-inspired inference-only control [1,3].
B3	Fresh retrieval + direct claim-aware judge.	SAFE-style search control, adapted to the same corpus [4].
B4	Scientific extraction/retrieval/verification with rationale.	SciClaims-style control [5]; official implementation if feasible, otherwise documented adaptation.
V	Fresh retrieval + sealed evidence reading + judgment + scope reconciliation.	Proposed full system, with separate D0 audit.
O	Gold evidence supplied to the same reader/judge.	Diagnostic evidence-oracle; not a deployable competitor.

The primary 2 x 2 ablation

Cross **evidence access** (D0 versus independent retrieval) with **reading policy** (claim-visible versus neutral-question, evidence-first). Hold decomposition, underlying model and downstream label interface constant. This separates the effects of new evidence and information separation. Run the natural-budget comparison and a matched total-token/call-budget comparison; extra computation is a competing explanation.

Secondary ablations, in priority order

1. Remove the counterevidence query while keeping query count constant.
2. Remove the dependency/inference-link check on multi-hop items.
3. Compare raw versus calibrated probabilities without changing predictions unnecessarily.
4. Same-family versus cross-family generator/checker pairing on the held-out agentic set.
5. Original citations visible first versus revealed after sealing the independent record.
6. Human study: same verdict/score with versus without the explanation.

Reproduce full MARCH training only as a separate stretch project with sufficient compute. The official repository provides training instructions and linked data; verify runnable artifacts and licenses in week 1. A prompt-only condition must never be labeled "MARCH reproduced."¹²

1. Li et al.. MARCH: Multi-Agent Reinforced Self-Check for LLM Hallucination. 2026, v1.

3. Dhuliawala et al.. Chain-of-Verification Reduces Hallucination in Large Language Models. 2023 preprint; 2024 publication.

4. Wei et al.. Long-form factuality in large language models. 2024.

5. Ortega and Gomez-Perez. SciClaims: An End-to-End Generative System for Biomedical Claim Analysis. EMNLP System Demonstrations, 2025.

12. Qwen-Applications. MARCH official implementation. repository, accessed 12 September 2026.

11. Metrics, statistics and result tables

Question	Primary measure	Supporting measures
Verification quality	Four-way macro-F1 on custom test; native scores on public benchmark.	Per-class precision/recall; confusion matrix; claim coverage; evidence selection scores.
False endorsement	Predicted supported among gold non-supported / all gold non-supported.	Wrong supported predictions / all supported predictions; report both denominators.
Evidence acquisition	Recall@k on judged evidence; newly found relevant/refuting documents.	Precision@k, D0/D1 overlap, unjudged rate and corpus-access failures.
Calibration	Multiclass Brier score: mean sum of $(p_{\text{class}} - \text{indicator})^2$.	NLL; reliability plots; ECE with bin settings; calibration sample size.
Selective use	Error among accepted claims versus fraction accepted.	Coverage at chosen threshold; false endorsements; abstention by reason.
Deployment	Median and p95 end-to-end latency, tokens and failures.	Warm/cold cache results; peak memory; cost per report and claim.

Statistical analysis

Preregister one primary comparison: V versus B2 for false endorsement on the custom held-out set. Use the factorial comparison to interpret the mechanism. Report effect sizes and 95% intervals even for null results. Bootstrap by question family, carrying both generator reports and all their claims together; 2,000 resamples is a proposed default. Twenty test families will produce limited precision.

Public-benchmark resampling should preserve source-document or question clusters where applicable. Use paired comparisons on the same items. Correct a declared family of secondary significance tests, for example with Holm adjustment, or label them exploratory. Repeated stochastic runs are not new independent test examples.

Experiment ledger and tables

T1: split counts, label distribution and annotation agreement.

T2: each method, native benchmark score, custom macro-F1, false endorsement and claim coverage.

T3: 2 x 2 ablation with budget columns and paired effect intervals.

T4: generator/checker transfer matrix.

T5: user-study condition effects and uncertainty.

Figures: architecture, reliability plot, risk-coverage curve, accuracy-cost trade-off and failure breakdown.

Freeze common extracted claims for the primary comparison; humans map them to reference spans. Report matched-claim classification and extraction coverage separately. Score end-to-end coverage against human claims; omissions are uncovered, not correct abstentions. Count infrastructure failures and show all-submitted-item results; never silently drop hard cases.

12. Software verification and error analysis

Tests should protect the scientific intervention, data isolation and evidence fidelity. They are more valuable than broad tests that simply repeat implementation details.

Test group	Concrete checks and expected behavior
Information isolation	Capture role payloads; asserted answers, full report, generator identity and D0 citations must not enter blind gatherer/reader inputs. Use canary strings and inspect semantic query leakage.
Scientific semantics	Negation, uncertainty, comparison, units, species, population and time remain intact. Human spot-check decomposition against source spans.
Evidence integrity	Every quotation matches the indexed snapshot and offsets. Duplicate papers are not counted as independent studies. Missing full text is surfaced.
Four-way distinctions	No relevant evidence -> unaddressed; unresolved relevant disagreement -> underdetermined; explicit reversal -> contradicted; scoped entailment -> supported.
Runtime and caching	Timeout, empty search, malformed output, overflow and retries are logged. Max rounds/calls enforced. Changed prompts/models invalidate cache.
Dataset isolation	No label files or held-out annotations accessible to model clients. Same question family cannot cross splits. Evaluation pools remain private.
Untrusted documents	Paper text containing instructions cannot call tools or alter role policy. Treat it as evidence data; check citation spoofing and corrupted metadata.
Renderer	No unsupported study-quality claims, missing citations, hidden uncertainty or partial-coverage endorsement in the final card.

Error taxonomy for manual review

Code missed claim, wrong scope, retrieval miss, ranking miss, evidence reading error, contradictory evidence mishandled, invalid synthesis, confidence error, explanation error and infrastructure failure. Review a predeclared random sample of at least 30 test errors if available; if fewer occur, review all. Also sample successful cases to detect accidental correctness.

For each error store the question, exact claim, D0/D1, gold spans, prediction, cause, plausible fix and whether the fix was evaluated. Do not tune on these test findings and then overwrite the original result. Any subsequent revision needs a new holdout or must be reported as post-test analysis.

Release gate

Require schema-valid outputs or explicit failure records, verified evidence links, no known information-boundary leak and reproducible aggregate scores from saved traces. Report performance weaknesses without changing these engineering acceptance criteria to chase a positive finding.

13. Non-expert trust-calibration study

Treat the two-month study as exploratory unless a prospective power analysis supports stronger claims. Prior work already studies confidence, explanations and human trust; the new question concerns this scientific validator and its specific evidence interface.¹¹

Design and participants

Target 30 non-expert adults, with 24-36 as a feasible recruitment range, plus 4-6 separate pilot participants. Prepare 12 short, expert-reviewed items that retain the scientific qualifiers and span supported, contradicted and unresolved evidence. Each participant sees each item once. Counterbalance condition assignment across people so item difficulty is not confounded with interface condition.

A: conclusion plus original citations.

B: A plus validator verdict and uncertainty/score.

C: B plus the evidence-linked explanation. B and C must use the same system verdict and confidence. Use four items per condition per participant, randomize order and record assignment. Across participants, rotate each item through all three conditions.

Per-item procedure and outcomes

Collect an initial 0-100 judgment that the conclusion is adequately supported, then show the assigned interface and collect the judgment again. Also collect a four-way evidence-status choice, one comprehension question, a use/defer decision and response time. For the binary probability outcome, define expert gold support as supported versus not established; do not describe "not established" as factually false.

Primary exploratory outcome: post-interface Brier score for evidence support, comparing C with B while adjusting for pre-interface belief and item effects. Also report appropriate acceptance of supported claims, rejection/deferral of unsupported claims, four-way accuracy and overreliance when the validator is wrong. Satisfaction and self-reported trust are secondary.

Wrong advice and ethics

Include naturally incorrect validator outputs in a prespecified error stratum where possible; report its sampling separately from natural error prevalence. If intentionally manipulated advice is necessary, require the appropriate ethics review and debriefing. Obtain informed consent and applicable institutional approval before recruitment; retain minimal personal data. Do not use clinical decisions or personalized health scenarios.

Analyze participant and item dependence with a mixed-effects model if supported by the sample; otherwise report participant-level paired summaries and uncertainty, explicitly limiting item generalization. Do not count 360 item responses as 360 independent participants. If recruitment or review is delayed, deliver the technical paper plus a preregistered study protocol and remove claims of demonstrated human benefit.

11. Zhang, Liao and Bellamy. Effect of Confidence and Explanation on Accuracy and Trust Calibration in AI-Assisted Decision Making. FAT*, 2020.

Week 1: Scope, reading and feasibility

By day 7, the research question, domain, resources and paper outline are fixed. The goal is to discover dependencies before substantial implementation.

Day / lead time	Work and completion target
Day 1 - 2h	Read the verified MARCH paper and map its architecture and section order. Write a one-page problem statement with inputs, outputs and explicit exclusions.
Day 2 - 2h	Read the blame-localization paper and CoVe. Build a comparison sheet: information seen by each role, whether retrieval is new, training needs, evaluation target.
Day 3 - 2h	Read SAFE and SciClaims. Rewrite novelty around a measurable intervention. Draft RQ1-RQ4 and provisional primary outcome.
Day 4 - 2h	Inspect SciFact files, native scorer and access/license terms. Choose biomedical abstract scope and document conditions that would force a domain change.
Day 5 - 2h	Profile available inference hardware with a 10-item smoke run. Freeze two candidate model families and record exact revisions after checking availability.
Day 6 - 6h	Create repository skeleton, data manifest and paper outline. Draft annotation rubric and study protocol. Identify annotators, adjudicator and ethics-review route; estimate actual availability.
Day 7 - 2h	Review feasibility and cut scope. Choose the intended paper venue and verify its current requirements. Finalize the week-2 backlog.

Writing and deliverables

Introduction skeleton, related-work matrix, initial limitations, 40-question topic plan, resource sheet and benchmark manifest.

Exit gate and fallback

Gate: one runnable model, accessible core data and an annotation path. If independent biomedical annotation is unavailable, prioritize native SciFact evaluation and a smaller explicitly preliminary agentic set; do not claim expert-validated scientific trust.

Weekly time: 18 lead-researcher hours. Collaborator annotation and unattended compute are tracked separately. Use up to 2 contingency hours this week only for an actual blocker; otherwise preserve them for final revision.

Week 2: Baselines and reproducible data

By day 14, a report can be judged against its original evidence, scored automatically and reproduced from a recorded configuration.

Day / lead time	Work and completion target
Day 8 - 2h	Write and validate Run, Claim, Evidence and Judgment schemas. Separate execution status from scientific labels.
Day 9 - 2h	Build the abstract index and retrieval adapter. Preserve document identifiers, text offsets, corpus hashes and original citations.
Day 10 - 2h	Implement direct-judge B0 and atomic same-evidence B1. Test numeric, negation and population examples.
Day 11 - 2h	Implement factored D0 reader B2. Add payload isolation assertions so blind inputs cannot contain asserted answers.
Day 12 - 2h	Create split manifest and calibration reservation. Build native benchmark adapters and confirm scorer behavior on hand-checked fixtures.
Day 13 - 6h	Run a 30-50-item development smoke evaluation. Inspect invalid outputs, latency, tokens and evidence recall. Prepare the 40 family-grouped research questions and generation configuration.
Day 14 - 2h	Freeze the first baseline release. Draft problem formulation, implementation setup and dataset sections with actual counts.

Writing and deliverables

Replayable B0-B2, corpus index, split hashes, development score report, annotation instructions v1 and paper methods draft.

Exit gate and fallback

Gate: every submitted item has a prediction or explicit failure record; no label leakage; native scoring works. Fix retrieval identifiers and data integrity before improving prompts.

Weekly time: 18 lead-researcher hours. Collaborator annotation and unattended compute are tracked separately. Use up to 2 contingency hours this week only for an actual blocker; otherwise preserve them for final revision.

Week 3: Independent evidence and staged judgment

By day 21, the full information boundary works, and the primary factorial ablation can run on development items.

Day / lead time	Work and completion target
Day 15 - 2h	Implement neutral-question generation preserving population, comparator and outcome. Review ten examples for leading phrasing and hidden asserted answers.
Day 16 - 2h	Implement independent gatherer with query variants, deduplication and retrieval logs. Hide D0 and the original report.
Day 17 - 2h	Implement evidence-only reader and seal its record. Add final claim judgment as a separate request.
Day 18 - 2h	Implement D0 audit and reconciliation. Preserve contradictory scopes instead of returning one unexplained overall score.
Day 19 - 2h	Implement B3 and the four factorial conditions. Keep model, corpus and context budget comparable.
Day 20 - 6h	Run 50 development claims across factorial conditions. Inspect retrieval gains, query bias, evidence overlap and token cost. Tune only development settings.
Day 21 - 2h	Draft architecture and pseudocode. Freeze provisional defaults for claim count, query count, passages, retries and stopping.

Writing and deliverables

Independent evidence pipeline, four-condition experiment config, boundary tests, architecture figure draft and query-audit notes.

Exit gate and fallback

Gate: independent reader has no direct exposure to the asserted answer, and every evidence excerpt resolves to a snapshot. If fresh retrieval gains vanish at matched budgets, retain that observation and test whether any other contribution remains.

Weekly time: 18 lead-researcher hours. Collaborator annotation and unattended compute are tracked separately. Use up to 2 contingency hours this week only for an actual blocker; otherwise preserve them for final revision.

Week 4: Agentic data, labels and explanation cards

By day 28, natural report generation is complete, the annotation rubric is stable, and a reader-facing prototype can be piloted.

Day / lead time	Work and completion target
Day 22 - 2h	Generate and freeze 80 reports from two families across 40 pre-split questions. Check malformed outputs without filtering for correctness.
Day 23 - 2h	Run a 20-claim double-annotation pilot. Time evidence search and rating. Resolve disagreements and finalize scope/label rules.
Day 24 - 2h	Implement dependency-link checks and claim-to-source coverage. Distinguish an invalid inference from an unsupported premise.
Day 25 - 2h	Build structured explanation cards with traceable quotations. Add missing-access, partial-coverage and uncertainty displays.
Day 26 - 2h	Start independent annotation in split order: development, calibration, then sealed test evaluation. Schedule adjudication and track actual throughput.
Day 27 - 6h	Implement B4 as a documented scientific-pipeline baseline. Pilot the user interface with 4-6 people only after applicable review/consent. Record comprehension failures and timing.
Day 28 - 2h	Review annotation feasibility. Finalize technical methods text and preregistered experiment plan; resolve any primary-outcome ambiguity.

Writing and deliverables

Frozen reports and trace manifest, human reference-claim inventories, rubric v2, user-study prototype, pilot notes and complete method draft.

Exit gate and fallback

Gate: the label distinction is annotatable and the main study fits the time available. If scientific judgments take twice the estimate, reduce the custom set prospectively or add collaborator time; preserve split integrity and disclose the reduction.

Weekly time: 18 lead-researcher hours. Collaborator annotation and unattended compute are tracked separately. Use up to 2 contingency hours this week only for an actual blocker; otherwise preserve them for final revision.

Week 5: Calibration and experiment freeze

By day 35, model/prompt choices are fixed, calibration is complete and all compared systems are ready for untouched evaluation.

Day / lead time	Work and completion target
Day 29 - 2h	Inspect development annotation and errors. Make the last permitted prompt/ranking changes and log the rationale.
Day 30 - 2h	Fit the simplest defensible calibrator on calibration data only. Save raw scores, fitted parameters and class counts.
Day 31 - 2h	Choose abstention thresholds using calibration data. Evaluate raw versus calibrated reliability without accessing test labels.
Day 32 - 2h	Finish native benchmark mapping and response-level RAGTruth adapter. Verify outside evidence is disabled for context-faithfulness scores.
Day 33 - 2h	Run software isolation, corrupted-evidence and runtime regression tests. Freeze code commit, prompts, corpus snapshot and experiment configurations.
Day 34 - 6h	Run the primary single-family experiment suite on held-out items. Save outputs before evaluator evidence pooling and adjudication. Begin matched-budget suite if throughput allows.
Day 35 - 2h	Check only operational completeness, not outcome-driven tuning. Finalize setup, metrics and annotation sections; confirm participant scheduling.

Writing and deliverables

Immutable experiment release, calibration model and policy, frozen prediction files, analysis scripts and study condition allocation.

Exit gate and fallback

Gate: all methods use the declared resources and evaluation scopes. If a test-time bug materially changes behavior, version the correction and rerun every affected comparison; never quietly replace only the weakest result.

Weekly time: 18 lead-researcher hours. Collaborator annotation and unattended compute are tracked separately. Use up to 2 contingency hours this week only for an actual blocker; otherwise preserve them for final revision.

Week 6: Benchmarks, ablations and system costs

By day 42, the technical evidence is largely complete and every main result has a reproducible table or figure.

Day / lead time	Work and completion target
Day 36 - 2h	Complete main runs and scorer validation. Separate native benchmark metrics from custom four-way results.
Day 37 - 2h	Run the matched-budget 2 x 2 ablation and counterevidence-query control. Include consumed tokens and calls in every comparison.
Day 38 - 2h	Run evidence-oracle diagnostics and dependency-check ablation. Attribute retrieval versus reasoning failures cautiously.
Day 39 - 2h	Run second-family checker transfer on the agentic test set. Reuse frozen reports and corpus; record all unavailable combinations.
Day 40 - 2h	Measure warm/cold latency, failures, memory and token totals. Plot risk-coverage and reliability using the already fixed calibration policy.
Day 41 - 6h	Finish adjudication, then compute paired cluster-bootstrap intervals. Create T1-T4 and select successes/failures using a declared sampling rule.
Day 42 - 2h	Write main results, ablations and computational analysis. Review whether the data actually supports each contribution.

Writing and deliverables

Main result tables, ablation results, paired intervals, reliability and risk-coverage figures, cost report and technical results draft.

Exit gate and fallback

Gate: denominators, splits, failure rates and budget differences are explicit. If primary gains are inconclusive, frame the result as limited or null; do not expand the test set after seeing results without declaring a new study.

Weekly time: 18 lead-researcher hours. Collaborator annotation and unattended compute are tracked separately. Use up to 2 contingency hours this week only for an actual blocker; otherwise preserve them for final revision.

Week 7: Human study and integrated draft

By day 49, the complete paper draft exists and the exploratory human evidence is analyzed if recruitment succeeds.

Day / lead time	Work and completion target
Day 43 - 2h	Freeze 12 expert-reviewed study items, condition texts and randomization. Verify B/C have identical verdicts and scores.
Day 44 - 2h	Begin participant sessions or monitored self-guided collection. Check consent, randomization and data integrity without changing outcome definitions.
Day 45 - 2h	Continue collection. Monitor recruitment and technical failures; log prespecified exclusions and all withdrawals.
Day 46 - 2h	Run the planned error analysis and review explanation faithfulness. Classify failures without modifying frozen system predictions.
Day 47 - 2h	Close the planned study window. Analyze participant-level outcomes and item dependence; report uncertain effects and validator-error strata.
Day 48 - 6h	Assemble the full manuscript, including human study, discussion, limitations and appendices. Link every table to a script and saved run.
Day 49 - 2h	Get an external technical read and a non-expert readability read. Ask specifically about novelty, comparison fairness and unsupported conclusions.

Writing and deliverables

Complete draft v1, study results or clearly labeled protocol-only appendix, failure taxonomy and reviewer issue list.

Exit gate and fallback

Gate: observed human benefit is claimed only if actually measured. If there are too few participants or no approved study, retain the technical paper and move human evaluation to future work.

Weekly time: 18 lead-researcher hours. Collaborator annotation and unattended compute are tracked separately. Use up to 2 contingency hours this week only for an actual blocker; otherwise preserve them for final revision.

Week 8: Revision and reproducibility package

By day 56, the paper and research artifact are ready for an external submission decision.

Day / lead time	Work and completion target
Day 50 - 2h	Resolve major reviewer questions. Narrow contribution statements to measured results and distinguish adaptations from reproductions.
Day 51 - 2h	Reproduce aggregate tables from saved outputs in a clean environment. Check rounding, seeds, split IDs and counts.
Day 52 - 2h	Complete appendix prompts, schemas, corpus access instructions, dataset card, annotation protocol and human-study materials.
Day 53 - 2h	Refresh related work and check exact source versions. Verify citation support, dataset licenses and current venue instructions.
Day 54 - 2h	Finalize abstract, introduction and conclusion. Check that all three use the same scope and do not overstate human or cross-model generalization.
Day 55 - 6h	Package clean setup instructions, a small runnable example, configs, evaluation scripts and permitted data. Render the paper and inspect all pages, figures and references.
Day 56 - 2h	Run the final checklist. Tag the final research release, write known limitations and choose submit, revise, or release as a preliminary technical report.

Writing and deliverables

Complete manuscript PDF/source, reproducibility package, source/license register, final result ledger and prioritized follow-up experiments.

Exit gate and fallback

Gate: no missing citations, unsupported headline numbers, inaccessible essential evaluation artifacts or undocumented changes. Submission remains a separate concrete action; the plan itself does not submit anything.

Weekly time: 18 lead-researcher hours. Collaborator annotation and unattended compute are tracked separately. Use up to 2 contingency hours this week only for an actual blocker; otherwise preserve them for final revision.

Resources, experiment size and scope cuts

Human time is the main constraint

Resource	Planning allowance	Interpretation
Lead researcher	144 scheduled + 16 contingency hours	18 hours/week plus optional 2-hour weekly buffer.
Scientific annotators	53 hours double labeling + 8-12 hours adjudication/search	400 claims x 4 minutes x 2 raters; replace estimate after pilot.
Study participants	30 x 20-25 minutes; 4-6 pilot participants separately	Recruitment/admin effort is additional; self-guided collection reduces scheduling load.
Independent review	2-4 collaborator hours	One methods review and one readability review, if available.

Compute ledger rather than guessed pricing

A sizing example: 80 reports x 5 claims x 6 conditions = 2,400 claim-condition evaluations. At an illustrative 3,000 input and 400 output tokens per evaluation, that is 7.2 million input and 0.96 million output tokens before retrieval, decomposition, extra samples and reruns. Staged conditions may consume much more; replace this approximation with week-2 measured totals.

For the primary checker, a planning suite of 6 methods on 300 public claims, about 200 custom test claims and 200 transfer responses yields about 4,200 item-condition evaluations. Agentic conditions may reuse decomposition only when that does not change the intervention. The transfer unit is a response, so token cost is not directly comparable to a claim. Cross-family and ablation runs are extra; do not multiply every dataset by every model automatically.

Compute cost = input tokens x input price + output tokens x output price + GPU hours x rental rate + retrieval fees. Use consistent price units. Local inference has electricity, memory and wall-time costs. Hardware fit depends on model, quantization, context and concurrency; verify by profiling rather than promising a specific GPU can run the full suite.

Optional participant-compensation budget: 30 x INR 300 = INR 9,000, plus INR 1,200-1,800 for 4-6 pilot participants at the same allowance. These are proposed budget allocations, not market rates. Scientific annotation compensation and compute require separate locally verified quotes.

Cut order if the schedule slips

First remove live retrieval, large-corpus stretch benchmarks and extra model families. Then reduce secondary ablations and transfer sample size prospectively. Preserve the primary factorial comparison, gold-label independence, failure accounting and evidence provenance. If the user study is infeasible, remove demonstrated-human-benefit claims. Full reinforcement-learning reproduction is outside the core 160-hour scope.

Paper and release acceptance checklist

Research claims

Every contribution maps to an experiment and a result table. Fresh retrieval, explanations and isolation are credited to relevant prior work. A hypothesis without sufficient evidence is presented as unresolved. "Model-agnostic" means a replaceable interface; any measured portability claim names the tested models. "Trust calibration" names whether it refers to model scores or people.

Method and data

Include full input/output schemas, role prompts, access boundaries, claim limits, query variants, retrieval settings, stopping rules and failure behavior. Publish exact splits and deduplication rules. Include annotation instructions, annotator qualifications, agreement before adjudication and the evidence-pooling procedure. Record abstract/full-text scope and which source material may be redistributed.

Results and figures

Every result has the number of questions, reports and claims, the unit of analysis, missing-output counts and uncertainty where appropriate. Separate native public-benchmark results from custom four-way results, and natural errors from controlled perturbations. Include token/call budgets in ablations. Show at least one unsupported-synthesis case and one validator failure alongside successes.

Reproduction package

Provide environment lockfile, model acquisition/revision instructions, corpus manifest, tiny example, batch runner, saved predictions, evaluation scripts, calibration parameters and figure scripts. Release source text only where allowed; otherwise provide IDs and acquisition instructions. Exclude credentials, participant identities and restricted source material. Include a clear license for original code and attribution for reused code.

Final decision rule

Observed outcome	Appropriate paper framing
Verification gain and credible human signal	A systems paper with technical results and an explicitly scoped exploratory human evaluation.
Verification gain, no human study	A verification systems paper with a proposed human-evaluation protocol; no demonstrated trust-improvement claim.
Human signal, weak technical gain	An interface/decision-support finding requiring careful error-stratum analysis and no claim of superior verification.
No clear gain over strong controls	A rigorous empirical or negative-results report; explain where complexity fails to justify its cost.

The first action after accepting this plan is week 1, day 1: create the comparison matrix and one-page problem definition. The most important early commitment is securing independent annotation; implementation speed cannot substitute for credible evaluation.

Sources

Primary sources used to ground the architecture comparisons, benchmark recommendations and evaluation cautions. Accessed 12 September 2026. The schedule, sample sizes, implementation defaults and budget examples are proposed project decisions. Paper-specific performance numbers are intentionally not used as targets.

1. MARCH: Multi-Agent Reinforced Self-Check for LLM Hallucination

Li et al.. 2026, v1. <https://arxiv.org/html/2603.24579v1>

Used for the section-order mapping and the distinction between an isolated checking pattern and full shared-policy training.

2. Who is the Agent to Blame? Localizing Faithfulness and Citation Mistakes in Agentic Deep Research

Hirsch et al.. 2026, v1. <https://arxiv.org/html/2608.24306v1>

Used to distinguish trace-based developer diagnosis from evaluating final conclusions for end users.

3. Chain-of-Verification Reduces Hallucination in Large Language Models

Dhuliawala et al.. 2023 preprint; 2024 publication. <https://arxiv.org/abs/2309.11495>

Used for independent question answering as an established verification control.

4. Long-form factuality in large language models

Wei et al.. 2024. <https://arxiv.org/abs/2403.18802>

Used to identify prior search-augmented atomic-fact evaluation and motivate a strong search baseline.

5. SciClaims: An End-to-End Generative System for Biomedical Claim Analysis

Ortega and Gomez-Perez. EMNLP System Demonstrations, 2025.

<https://aclanthology.org/2025.emnlp-demos.11/>

Used to identify an existing scientific verification system with natural-language explanations. Published version is the preferred citation.

6. Fact or Fiction: Verifying Scientific Claims

Wadden et al.. EMNLP, 2020. <https://aclanthology.org/2020.emnlp-main.609/>

Used to choose scientific claim/evidence verification and preserve native benchmark scoring.

7. SciFact-Open: Towards open-domain scientific claim verification

Wadden et al.. 2022. <https://arxiv.org/abs/2210.13777>

Used to identify a larger-corpus optional retrieval evaluation, with incomplete-judgment cautions in the proposed protocol.

Sources continued

8. RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models

Niu et al.. ACL, 2024. <https://aclanthology.org/2024.acl-long.585/>

Used for a separate context-faithfulness transfer check; its annotations are not four-way scientific truth labels.

9. Retromorphic Testing with Hierarchical Verification for Hallucination Detection in RAG

Yu et al.. 2026, v2. <https://arxiv.org/abs/2603.27752v2>

Used to credit claim-level context checking and evidence-linked auditing. Consult v2 rather than relying on the earlier summary.

10. On Calibration of Modern Neural Networks

Guo et al.. ICML, 2017. <https://proceedings.mlr.press/v70/guo17a.html>

Used for the distinction between raw confidence and held-out calibrated probabilities.

11. Effect of Confidence and Explanation on Accuracy and Trust Calibration in AI-Assisted Decision Making

Zhang, Liao and Bellamy. FAT*, 2020. <https://arxiv.org/abs/2001.02114>

Used to avoid a false claim that human trust calibration has never been studied and to motivate measuring behavior separately from trust ratings.

12. MARCH official implementation

Qwen-Applications. repository, accessed 12 September 2026.

<https://github.com/Qwen-Applications/MARCH>

Used to distinguish a training reproduction from an inference-only adaptation. Repository instructions were inspected; no training or benchmark reproduction was executed for this plan.

13. SciFact data, models and evaluation code

Allen Institute for AI. repository, accessed 12 September 2026.

<https://github.com/allenai/scifact>

Used for the planned data/scorer access route. Data download, exact available split counts and runnable compatibility are week-1 implementation checks.

14. Missci: Reconstructing Fallacies in Misrepresented Science

Glockner et al.. ACL, 2024. <https://aclanthology.org/2024.acl-long.240/>

Used as an optional scientific misrepresentation/fallacy stress test, not as interchangeable four-way gold labels.